

Elektrotehnički fakultet u Beogradu

Specifikacija baze podataka

Projekat: BeachBooker

Predmet: Principi softverskog inženjerstva (SI3PSI)

Tim: Tropski bar

Verzija 1.0 · 21. maj 2026.

Istorija revizija

Revizija	Datum	Autor	Opis
1.0	21.05.2026.	Tim Tropski bar	Inicijalna verzija

Sadržaj

1. Uvod (namena, ciljne grupe, organizacija, rečnik, otvorena pitanja)
2. Model podataka (ER, IE dijagram, relacione šeme)
3. Tabele (10 tabela)
4. Prilog — SQL skripta

1. Uvod

1.1 Namena

Baza podataka za **BeachBooker** centralizuje podatke o korisnicima, beach barovima, ležaljka, rezervacijama i pratećim entitetima. Backend je **Django**; klijent koristi **HTML** i **AJAX** za dinamičke mape i rezervacije. Dokument sadrži ER prikaz, IE dijagram, relacione šeme i katalog tabela — osnova za implementaciju i odbranu projekta.

DBMS: MySQL 8+ · **Šema:** beachbooker

1.2 Ciljne grupe

Vođa tima, projektant baze, backend razvoj i ostali članovi tima — za usklađivanje imena polja, integraciju i održavanje modela.

1.3 Organizacija dokumenta

Poglavlje 2 — modeli; poglavlje 3 — tabele; prilog — SQL skripte u repozitorijumu (`schema.sql`).

1.4 Rečnik pojmova

Pojam	Značenje
ER / IE	Entity-Relationship / Information Engineering notacija

PK / FK	Primarni / strani ključ
Gost	Neregistrovani posetilac — nema reda u bazi
Dostupnost	Izvedeno iz rezervacija, nema posebne tabele

1.5 Otvorena pitanja

Br.	Problem	Rešenje
1	Više barova po vlasniku?	Da u šemi (1,n); MVP UI može ograničiti na jedan.
2	Bundle / review u MVP UI?	Tabele postoje; UI kasnije.

2. Model podataka

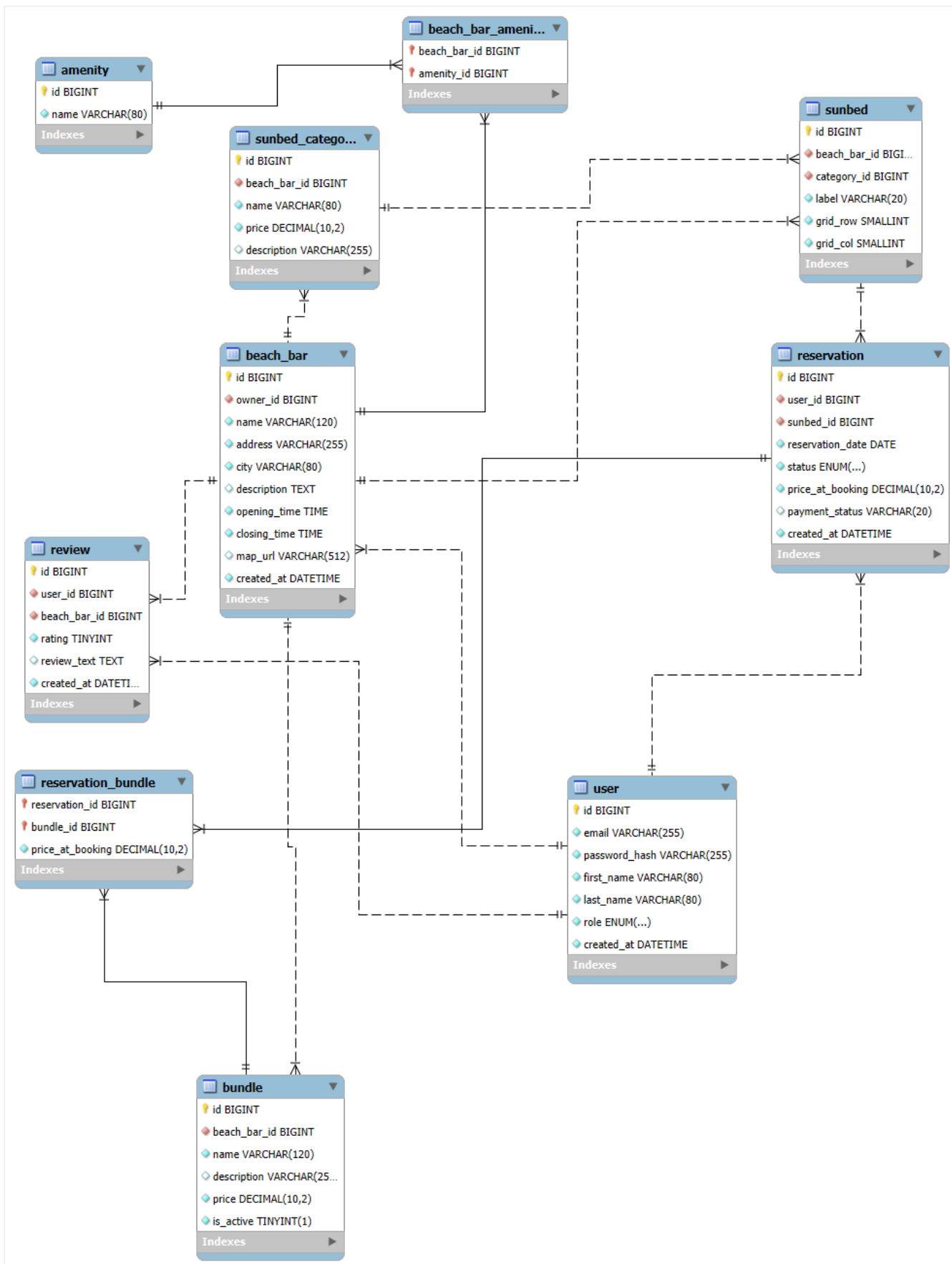
2.1 Dijagram ER notacije

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USER (1,n) BEACH_BAR · USER (1,n) RESERVATION · USER (1,n) REVIEW
BEACH_BAR (1,n) SUNBED_CATEGORY · (1,n) SUNBED · (1,n) BUNDLE
BEACH_BAR (n,m) AMENITY preko BEACH_BAR_AMENITY
SUNBED (1,n) RESERVATION · RESERVATION (n,m) BUNDLE preko RESERVATION_BUNDLE
Jedinstveno: (beach_bar_id, label) na SUNBED; (sunbed_id, reservation_date) na
RESERVATION

```

2.2 Dijagram IE notacije



Slika 1. Logički model baze podataka (MySQL Workbench). Izvor: beach_booker_ie.png, model: beachbooker_EER_model.mwb

2.3 Šema relacione baze podataka

```

user (id, email, password_hash, first_name, last_name, role, created_at)
beach_bar (id, owner_id, name, address, city, description, opening_time, closing_time,
map_url, created_at)
amenity (id, name)
beach_bar_amenity (beach_bar_id, amenity_id)
sunbed_category (id, beach_bar_id, name, price, description)
sunbed (id, beach_bar_id, category_id, label, grid_row, grid_col)
reservation (id, user_id, sunbed_id, reservation_date, status, price_at_booking,
payment_status, created_at)
bundle (id, beach_bar_id, name, description, price, is_active)
reservation_bundle (reservation_id, bundle_id, price_at_booking)
review (id, user_id, beach_bar_id, rating, review_text, created_at)

```

3. Tabele

3.1 user

Nalozi: registrovani korisnik, vlasnik, administrator. Lozinka: password_hash.

Name	Datatype	PK	FK	Napomene
id	BIGINT UNSIGNED	Yes	No	AUTO_INCREMENT
email	VARCHAR(255)	No	No	UNIQUE, NOT NULL
password_hash	VARCHAR(255)	No	No	NOT NULL
first_name	VARCHAR(80)	No	No	NOT NULL
last_name	VARCHAR(80)	No	No	NOT NULL
role	ENUM	No	No	registered owner admin
created_at	DATETIME	No	No	DEFAULT CURRENT_TIMESTAMP

3.2 beach_bar

Name	Datatype	PK	FK	Napomene
id	BIGINT UNSIGNED	Yes	No	AUTO_INCREMENT

owner_id	BIGINT UNSIGNED	No	user.id	NOT NULL
name	VARCHAR(120)	No	No	NOT NULL
address	VARCHAR(255)	No	No	NOT NULL
city	VARCHAR(80)	No	No	NOT NULL
description	TEXT	No	No	NULL
opening_time	TIME	No	No	NOT NULL
closing_time	TIME	No	No	NOT NULL
map_url	VARCHAR(512)	No	No	NULL
created_at	DATETIME	No	No	

3.3 amenity

Name	Datatype	PK	FK	Napomene
id	BIGINT UNSIGNED	Yes	No	AUTO_INCREMENT
name	VARCHAR(80)	No	No	UNIQUE, NOT NULL

3.4 beach_bar_amenity

Name	Datatype	PK	FK	Napomene
beach_bar_id	BIGINT UNSIGNED	Yes	beach_bar.id	
amenity_id	BIGINT UNSIGNED	Yes	amenity.id	

3.5 sunbed_category

Name	Datatype	PK	FK	Napomene
id	BIGINT UNSIGNED	Yes	No	
beach_bar_id	BIGINT UNSIGNED	No	beach_bar.id	NOT NULL
name	VARCHAR(80)	No	No	NOT NULL
price	DECIMAL(10,2)	No	No	NOT NULL
description	VARCHAR(255)	No	No	NULL

3.6 sunbed

Name	Datatype	PK	FK	Napomene
id	BIGINT UNSIGNED	Yes	No	
beach_bar_id	BIGINT UNSIGNED	No	beach_bar.id	UNIQUE sa label
category_id	BIGINT UNSIGNED	No	sunbed_category.id	
label	VARCHAR(20)	No	No	npr. P3
grid_row	SMALLINT UNSIGNED	No	No	mapa
grid_col	SMALLINT UNSIGNED	No	No	mapa

3.7 reservation

Name	Datatype	PK	FK	Napomene
id	BIGINT UNSIGNED	Yes	No	
user_id	BIGINT UNSIGNED	No	user.id	
sunbed_id	BIGINT UNSIGNED	No	sunbed.id	UNIQUE sa reservation_date
reservation_date	DATE	No	No	NOT NULL
status	ENUM	No	No	active completed cancelled
price_at_booking	DECIMAL(10,2)	No	No	snapshot
payment_status	VARCHAR(20)	No	No	NULL
created_at	DATETIME	No	No	

3.8 bundle

Name	Datatype	PK	FK	Napomene
id	BIGINT UNSIGNED	Yes	No	
beach_bar_id	BIGINT UNSIGNED	No	beach_bar.id	
name	VARCHAR(120)	No	No	
description	VARCHAR(255)	No	No	NULL
price	DECIMAL(10,2)	No	No	
is_active	TINYINT(1)	No	No	DEFAULT 1

3.9 reservation_bundle

Name	Datatype	PK	FK	Napomene
reservation_id	BIGINT UNSIGNED	Yes	reservation.id	
bundle_id	BIGINT UNSIGNED	Yes	bundle.id	
price_at_booking	DECIMAL(10,2)	No	No	NOT NULL

3.10 review

Name	Datatype	PK	FK	Napomene
id	BIGINT UNSIGNED	Yes	No	
user_id	BIGINT UNSIGNED	No	user.id	
beach_bar_id	BIGINT UNSIGNED	No	beach_bar.id	
rating	TINYINT UNSIGNED	No	No	CHECK 1–5
review_text	TEXT	No	No	NULL
created_at	DATETIME	No	No	

4. Prilog

SQL: `schema.sql, schema_forward.sql` · Model: `beachbooker_EER_model.mwb`